increase the bit-rate as soon as it detected that there were voices speaking. cdmaOne vocoders encode voice at 9600 bits per second. After applying the pseudo-random code, the signal is spread to 1.23 Mbps.

As mentioned before, CDMA was first tinkered with by the US Military. Because messages are spread out over a band of frequencies, an eavesdropper would dismiss this signal as electrical noise. Even if this eavesdropper did identify the signal as a CDMA transmission, it would be very difficult for him to decode it.

The capacity of a CDMA network is about four to five times as much as that of a GSM network. It offers better coverage, better call quality and practically unbreakable privacy. Because all calls on a CDMA network use the same frequency band, there is no need for 'frequency planning' in a CDMA network - all cells work on the same frequency.

We now move on to more advanced things, having dealt with the first and second generation technologies.

The General Packet Radio Service - GPRS

GPRS is a data transfer method that integrates neatly with GSM networks. It employs unused time slots in the TDMA channels to transfer data. This is a lot faster than the method previously mentioned, of circuit-switched data transfer.

We should, perhaps, look into the differences between circuit switching and packet switching before we start making claims about which is better.

In a typical network, there are a number of different paths that could be employed to establish a link between two points. A *network controller* selects the best path, and once this path is established, communication can begin. Packets of data are sent by the transmitter; they travel the network along this path to reach the receiver. This is called *Circuit Switching*. "Quite good", you might